New York University Tandon School of Engineering
Computer Science and Engineering

CS-GY 9223D: Midterm Practice.

Practice Problems

Random variables and concentration.

1. Show that for any random variable X, $\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$. $\mathbb{E}[X^2] - \mathbb{E}[X]^2 = \text{Var}[X]$, and variance must be non-negative.

2. Show that for independent X and Y with $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, $\text{Var}[X \cdot Y] = \text{Var}[X] \cdot \text{Var}[Y]$. Since X and Y are independent $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] = 0$. So $\text{Var}[XY] = \mathbb{E}[X^2Y^2] = \mathbb{E}[X^2]\mathbb{E}[Y^2]$. The last step follows from that fact that X^2 and Y^2 are independent because X and Y are. Finally, since $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, we have $\mathbb{E}[X^2] = \text{Var}[X]$ and $\mathbb{E}[Y^2] = \text{Var}[Y]$. Plugging in gives the proof.

3. Given a random variable X, can we conclude that $\mathbb{E}[1/X] = 1/E[X]$? If so, prove this. If not, give an example where the equality does not hold. Not true. Try a $\{-1, 1\}$ random variable. $E[1/X]$ is not infinite.

4. Indicate whether each of the following statements is **always** true, **sometimes** true, or **never** true. Provide a short justification for your choice.

   (a) $\Pr[X = s \text{ and } Y = t] > \Pr[X = s]$. ALWAYS SOMETIMES **NEVER**

   (b) $\Pr[X = s \text{ or } Y = t] \leq \Pr[X = s] + \Pr[Y = t]$. **ALWAYS** SOMETIMES NEVER (By union bound.)

   (c) $\Pr[X = s \text{ and } Y = t] = \Pr[X = s] \cdot \Pr[Y = t]$. ALWAYS **SOMETIMES** NEVER

5. Assume there are 1000 registered users on your site $u_1, \ldots, u_{1000}$, and in a given day, each user visits the site with some probability p_i. The event that any user visits the site is independent of what the other users do. Assume that $\sum_{i=1}^{1000} p_i = 500$.

   (a) Let X be the number of users that visit the site on the given day. What is $E[X]$? By linearity of expectation, 500.

   (b) Apply a Chernoff bound to show that $Pr[X \geq 600] \leq .01$. Setting $\epsilon = .2$ and applying the bound from class we have $Pr[X \geq 600] \leq e^{-.2^2 500/2.2} = .0001$.

   (c) Apply Markovs inequality and Chebyshevs inequality to bound the same probability. How do they compare? By Markov's $Pr[X \geq 600] \leq \frac{1}{1.2} = .833$. For Chebyshev's we need a variance calculation. One thing we can say is that $\text{Var}[X] = \sum_{i=1}^{1000} \text{Var}[p_i] \leq \sum_{i=1}^{1000} p_i = 500$. By Chebyshev, $\Pr[|X - 500| \geq k\sqrt{500}] \leq \frac{1}{k^2}$. Setting $k = 100/\sqrt{500} = 4.47$ gives $\frac{1}{k^2} = .05$.

6. Give an example of a random variable and a deviation t where Markovs inequality gives a tighter bound than Chebyshevs inequality. Take a uniform $\{0, 1\}$ random variable and bound $\Pr[X \geq 1]$. Markov's will give the tight 1/2 bound, but Chebyshev's gives a vacuously weak bound.

Hashing, Dimensionality Reduction, High Dimensional Vectors

1. SimHash can be applied to binary vectors $y, q \in \{0, 1\}^d$. How does doing so compare to MinHash? How is the Jaccard similarity $J(y, q)$ related to the cosine similarity $\cos(\theta(y, q))$?

2. Suppose there is some unknown vector $\boldsymbol{\mu}$. We receive noise perturbed random samples of the form $\mathbf{Y}_1 = \boldsymbol{\mu} + \mathbf{X}_1, \ldots, \mathbf{Y}_k = \boldsymbol{\mu} + \mathbf{X}_k$ where each $\mathbf{X}_i$ is a random vector with each of its entries distributed as an independent random normal $\mathcal{N}(0, 1)$. From our samples $\mathbf{Y}_1, \ldots, \mathbf{Y}_k$ we hope to estimate $\boldsymbol{\mu}$ by $\tilde{\boldsymbol{\mu}} = \frac{1}{k} \sum_{i=1}^{k} \mathbf{Y}_i$.

(a) How many samples k do we require so that $\max|\boldsymbol{\mu}-\tilde{\boldsymbol{\mu}}| \le \epsilon$ with probability 9/10? If we can show that for all $i \in 1,\dots,n$, $|\boldsymbol{\mu}_i - \tilde{\boldsymbol{\mu}}_i| \le \epsilon$ with probability $1-\frac{1}{10n}$ then by a union bound we will have that $\max|\boldsymbol{\mu}-\tilde{\boldsymbol{\mu}}| \le \epsilon$ with probability $1-\frac{1}{10}$. So we focus on this simpler problem. Notice that $\tilde{\boldsymbol{\mu}}_i = \boldsymbol{\mu}_i + \frac{1}{k}\sum_{j=1}^{k}[\mathbf{X}_j]_i$, where $[\mathbf{X}_j]_i$ is the ith entry of $\mathbf{X}_j$, which is a norm random variable. Since the sum of random normals is norm, we have that $\frac{1}{k}\sum_{j=1}^{k}[\mathbf{X}_j] \sim \frac{1}{k}\mathcal{N}(0,k) = \mathcal{N}(0,1/k)$. Applying the Gaussian tail bound from lecture, we have thus have that $\Pr[|\boldsymbol{\mu}_i - \tilde{\boldsymbol{\mu}}_i| \ge \alpha/\sqrt{k}] \le e^{-O(\alpha^2)}$. Setting $\alpha = O(\sqrt{\log(1/(1/10n))} = O(\sqrt{\log n})$ and $k = O(\log n/\epsilon^2)$ gives the bound we need. So overall we need $k = O(\log n/\epsilon^2)$ samples.

(b) How many samples k do we require so that $\|\boldsymbol{\mu}-\tilde{\boldsymbol{\mu}}\|_2 \le \epsilon$ with probability 9/10? The proof is essentially the same. To ensure that $\|\boldsymbol{\mu}-\tilde{\boldsymbol{\mu}}\|_2 \le \epsilon$, it suffices to have $|\boldsymbol{\mu}_i - \tilde{\boldsymbol{\mu}}_i| \le \sqrt{\epsilon/n}$ for all i. This will happen with probability 9/10 as long as $k = O(\log n/(\sqrt{\epsilon/n})^2) = k = O(n\log n/\sqrt{\epsilon}^2)$.

3. Let $\boldsymbol{\Pi}$ be a random Johnson-Lindenstrauss matrix (e.g. scaled random Gaussians) with $O(\log(1/\delta)/\epsilon^2)$ rows. Prove that with probability $(1-\delta)$,

$$\min_{\mathbf{x}} \|\boldsymbol{\Pi}\mathbf{A}\mathbf{x} - \boldsymbol{\Pi}\mathbf{b}\|_2^2 \le (1+\epsilon)\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x}-\mathbf{b}\|_2^2$$

Let $\mathbf{x}^* = \arg\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x}-\mathbf{b}\|_2^2$. Then by the distributional Johnson-Lindenstrauss lemma we have $\|\boldsymbol{\Pi}\mathbf{A}\mathbf{x}^* - \boldsymbol{\Pi}\mathbf{b}\|_2^2 = \|\boldsymbol{\Pi}(\mathbf{A}\mathbf{x}^*-\mathbf{b})\|_2^2 \le (1+\epsilon)\|\mathbf{A}\mathbf{x}^*-\mathbf{b}\|_2^2$. And then we have that $\min_{\mathbf{x}} \|\boldsymbol{\Pi}\mathbf{A}\mathbf{x}-\boldsymbol{\Pi}\mathbf{b}\|_2^2 \le \|\boldsymbol{\Pi}\mathbf{A}\mathbf{x}^* - \boldsymbol{\Pi}\mathbf{b}\|_2^2$, which gives the result.

Is the following also true with high probability?

$$(1-\epsilon)\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x}-\mathbf{b}\|_2^2 \le \min_{\mathbf{x}} \|\boldsymbol{\Pi}\mathbf{A}\mathbf{x} - \boldsymbol{\Pi}\mathbf{b}\|_2^2$$

This statement is not true. The problem is that, if we let $\mathbf{y}^* = \arg\min_{\mathbf{x}} \|\boldsymbol{\Pi}\mathbf{A}\mathbf{x}-\boldsymbol{\Pi}\mathbf{b}\|_2^2$, we can't apply distributional JL lemma to show that $\|\boldsymbol{\Pi}\mathbf{A}\mathbf{y}^* - \boldsymbol{\Pi}\mathbf{b}\|_2^2 \approx \|\mathbf{A}\mathbf{y}^* - \mathbf{b}\|_2^2$ because $\mathbf{y}^*$ is *dependent* on $\boldsymbol{\Pi}$. JL lemma only holds to preserve the norms of vectors fixed ahead of time, that don't depend on $\boldsymbol{\Pi}$.

Convex optimization

From *Convex Optimization* book (`https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf`):

- **Exercises:** 3.7, 3.10, 3.11 (first part), 3.21 (lots of other problems if you want more practice, but many are on the harder si)

1. Let $f_1(x),\dots,f_n(x)$ be β-smooth convex functions and let $g(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$ be their average. Show that g is β-smooth.

$$\begin{aligned}\|\nabla g(x) - \nabla g(y)\|_2 &= \frac{1}{n}\|\sum_{i=1}^{n}\nabla f_i(x) - \nabla_i(y)\|_2 \\ &\le \frac{1}{n}\sum_{i=1}^{n}\|\nabla f_i(x) - \nabla_i(y)\|_2 \\ &\le \frac{1}{n}\sum_{i=1}^{n}\beta\|x-y\|_2 = \beta\|x-y\|_2\end{aligned}$$

The first inequality is from triangle inequality, the second is from assumed smoothnes of $f_1,\dots,f_n$.

2. Let $f:\mathbb{R}\to\mathbb{R}$ be a β-smooth, α-strongly convex function. Let $g(x) = f(c\cdot x)$ for some constant $0 < c < 1$. How does g's smoothness and strong convexity compare to that of f? How about g's condition number? For a scalar function, $\beta = \max_x f''(x)$. $g''(x) = c^2 f''(cx)$. $\max_x f''(x) = \max_x f''(cx)$, so g's smoothness parameter is larger than that of f by a factor of c by a factor c^2. By the same argument, and using that $\beta = \min_x f''(x)$, we get that g's strong convexity parameter α is also larger by a factor of c^2. The condition number is the ratio between β an α. Since the c^2 factors will cancel out, it stays the same.

3. Let $f(x) = x^4$. Is f G-Lipschitz for finite G? Is f β-smooth for finite B? For f to be G-Lipschitz, it gradient, in this case $4x^3$, would need to bounded in magnitude by some fixed constant. In this case it is not, because $4x^3 \to \infty$ as $x \to \infty$. Similarly, for it to be β smooth, the second derivative, $12x^2$ would need to be bounded. But again it isn't. So the answer is no for both.